

Suggested implementation project plan-2 for NIT-Trichy course: *Computer Networks*

Project on video streaming over UDP networks: project duration (27 Aug.-31 Oct., 2025)

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We consider a rate adaptive end-to-end video transport system (BAVT) realized over a bandwidth-limited network path. The BAVT system is realized by a session-layer network algorithm, known as AIMD (additive increase multiplicative decrease). Figure 1 shows the architecture of a BAVT, with a delineation of the core adaptation functions to adjust the send rate and the augmented system management functions that exercise an external control. We provide below a control-theoretic management of the BAVT functionality.

Complexity induced by video flow characteristics

The packet flow from a video source is bursty, making its bandwidth demand highly unpredictable. If multiple video sources share the path, the statistical multiplexing effects also impact the bandwidth use. The lack of knowledge about bandwidth use is compounded by a presence of cross-traffic along the path, thereby making the available bandwidth B_{av}^* difficult to estimate [1, 2]. If λ is the rate at which the video source attempts to send packets, we denote the bandwidth demand of the flow as: $\mathcal{F}^*(\lambda, [m, \rho, \dots])$ — where m is the residual packet loss rate on the link and ρ depicts the burstiness of packet flow. With flow aggregation over the data path, ρ depends on the number flows admitted N and the traffic smoothing exercised on flows¹ [3]. Since neither B_{av}^* nor $\mathcal{F}^*$ is known in an exact form, the adaptation condition: $\mathcal{F}^*(\lambda', [m, \rho]) < B_{av}^*$, is determined by model-based control — where λ' is the sustainable send rate.

A model-based description of the raw transport sub-system embodied in the BAVT system may be given as:

$$L^* = (1 - \frac{B_{av}^*}{\mathcal{F}^*(\lambda, [m, \rho, \dots])}), \quad (1)$$

where L^* is the packet loss rate experienced due to congestion ($L^* > m$). In the model², B_{av}^* constitutes the internal state of end-to-end transport connection that forms the core of BAVT system and L^* depicts the system output result of sending video data over the connection.

In the absence of exact knowledge about $\mathcal{F}^*(\dots)$ and/or B_{av}^* , an 'additive increase multiplicative decrease' (AIMD) algorithm [4, 5] is employed as a coarse modeling approximation to adjust the send rate to λ' matching the available bandwidth B_{av}^* . The AIMD steps are:

- Set the new send rate as: $\lambda^{(\text{new})} = \lambda^{(\text{cur})} \times \exp(-\beta L)$ if $L > \delta_h$, where L is the observed end-to-end loss ratio and δ_h is a threshold to indicate congestion, and $\beta > 0$;
- Set the new send rate as: $\lambda^{(\text{new})} = \lambda^{(\text{cur})} + \alpha$ when $L < \delta_l$, where δ_l is a loss threshold below which congestion is deemed to be absent ($0 < \delta_l < \delta_h$), and $\alpha > 0$.

$L \in [\delta_l, \delta_h]$ depicts a stable operating region for the BAVT system, with the send rate λ' exhibiting a limit-cycle as determined by the hysteresis parameter ($\delta_h - \delta_l$) — where δ_l and δ_h are small loss percentages: say, 0.75% and 1% respectively. Note that a video packet delayed by longer than a threshold D is deemed as lost. Figure 2 shows the algorithmic schema of AIMD-based rate adaptation procedure. The³ AIMD

¹A measure of the burstiness of a flow is expressible in terms of the peak and average rates: $\rho = [1 - \frac{\lambda(a)}{\lambda(p)}]$, with $\rho = 0$ depicting a CBR flow (e.g., music) and $\rho \rightarrow 1.0(-)$ depicting a highly bursty flow.

²As a convention, '*' denotes the true value of a system parameter or function. Determining the true value is however subject to observational and/or modeling errors. A parameter or function without '*' denotes the actual value made available to the controller module with the use of finite resources by the underlying computation.

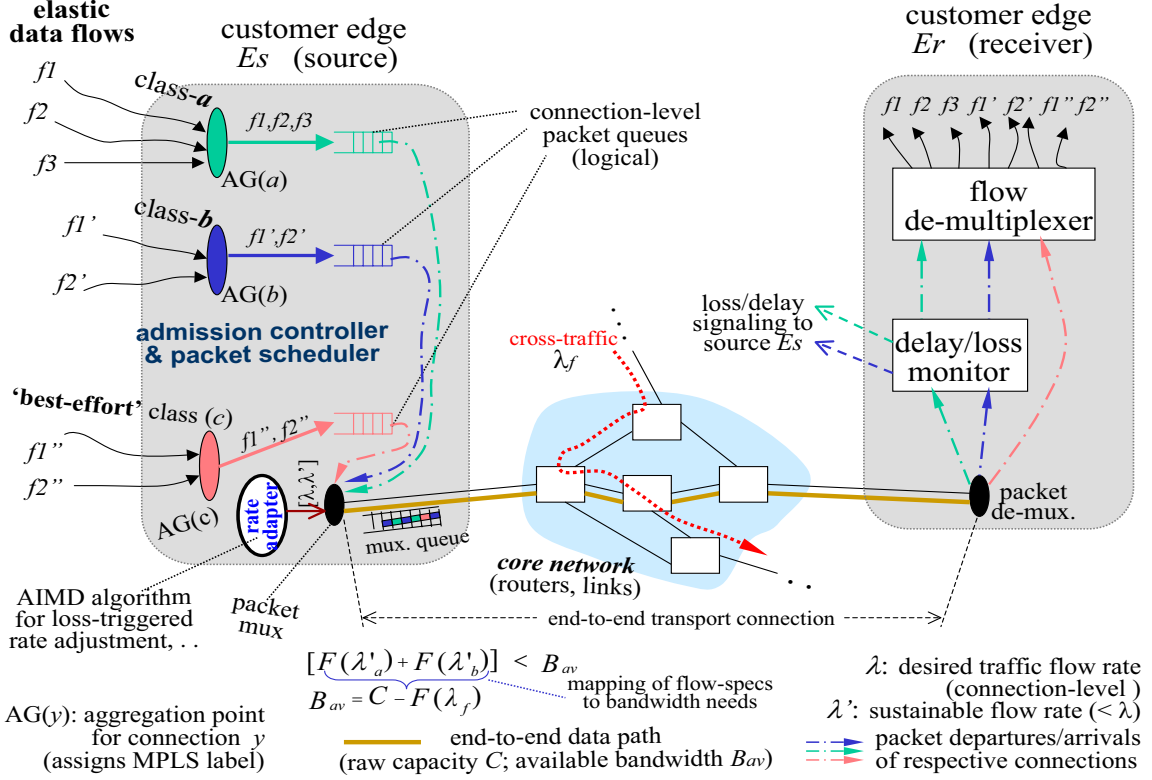


Figure 1: CPS-based view of bandwidth-adaptive video transport system

parameters β/α determine how fast the send rate can be slowed down or ramped up without causing a jitter in the user-level perceptual QoS. The control rounds are separated by a time interval determined by τ_{conn} : the time-constant of raw transport connection, i.e., the time needed for a rate reduction/increase to have an observable impact on the bandwidth use (say, for the existing packets in router queues to trickle out and for reporting the change in packet loss) [6].

Whether a rate reduction relieves the congestion or not can be ascertained only by exercising the rate reduction on the transport connection and observing the end-to-end packet loss rate L therein. If $L > \delta_h$, the under-reduction in send rate may be compensated in a subsequent estimation of rate reduction. The observe-and-adapt cycle continues over multiple rounds q until the congestion is relieved, i.e., $L < \delta_l$. The total time to adapt: $T_{cAIMD} \approx q\tau_{conn}|_{q \geq 1}$, depicts the convergence latency of rate adaptation process.

Black-box programming view of transport connection

The transport connection is abstracted as a computational function of the form: $L = \text{net}(\lambda)$, where $\lambda > 0$ and $L > 0$. The internal details of $\text{net}(\dots)$ to compute L for an input λ are not known to the programmer, i.e., $\text{net}(\lambda)$ appears as a black-box taking λ as input and returning L as output. The programmer however knows how the function $\text{net}(\dots)$ behaves when λ changes, as given by the monotonicity relationship:

$$\text{net}(\lambda + \Delta) > \text{net}(\lambda) > \text{net}(\lambda - \Delta) \quad (2)$$

for $\Delta > 0$. In other words, an increase in λ leads to a higher L , and a decrease in λ leads to a lower L .

Suppose λ_0 is an initial input for which $\text{net}(\lambda_0)$ returns a value $L_0 > \delta_l$, where $\delta_l > 0$. We develop an algorithmic procedure that reduces λ in multiple steps of $(\beta \times L)$ decrements to a value λ_f such that

³The hysteresis parameter $(\delta_h - \delta_l)$ prevents ping-pong effects on the rate control when the external conditions change fast.

Iterative procedure for rate adjustment

Additive Increase Multiplicative Decrease (AIMD)

In each ‘loss reporting’ interval

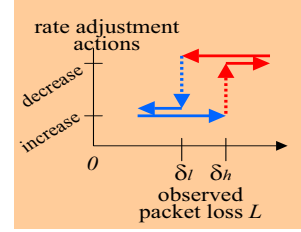
$$\lambda_{(new)} = \lambda_{(cur)} - \beta.L \quad \text{when } L > \delta_h, \\ \text{where } \beta > 0$$

$$\lambda_{(new)} = \lambda_{(cur)} + \alpha \quad \text{when } L < \delta_l, \\ \text{where } \alpha > 0$$

L : observed packet loss rate

δ_l, δ_h : Acceptable loss thresholds

[$\delta_l < \delta_h$ to avoid ping-pong effect]



^^ Each execution of this procedure constitutes a “**control iteration**”

^^ A sequence iterations that lead to a steady-state in bandwidth allocation (when the flow specs changes or a set of new flows are admitted) constitutes a “**control round**”

Figure 2: Schema of AIMD-based rate adaptation procedure in a BAVT system

$\text{net}(\lambda_f) < \delta_l$. When $L < \delta_l$, the procedure increases λ in multiple steps of the constant α to a value $\lambda_{f'}$ such that $\text{net}(\lambda_{f'}) > \delta_h$. Thereupon, the procedure switches back to the rate reduction cycle. The sustainable send rate is given by:

$$\lambda' = \frac{\lambda_f + \lambda_{f'}}{2},$$

where $0 < \lambda' < \lambda$. The repetitive computation steps in the program (i.e., iterations) look like the following:

```

L := net(λ)    /* Initial output L0 ≥ δl */
λ(g) := λ;
while (true)
  while (L ≥ δl) /* loop 1: multiplicative decrease */
    λ(g) := λ(g) - βL;
    L = net(λ(g));
  while (L < δh) /* loop 2: additive increase */
    λ(g) := λ(g) + α;
    L := net(λ(g));
  if (steady_state is reached)
    break;
λ' := λ(g).

```

Here, each iteration of **while** loop 1 realizes the multiplicative decrease phase of AIMD rule and that of loop 2 realizes the additive increase phase. Given the monotonicity property of $\text{net}(\dots)$ function, we argue that the program terminates, i.e., finds λ' in a finite number of iterations.

QoE aspects of video rate adaptation

The QoE of BAVT systems depends on multiple parameters that capture the perceptual QoS experienced by a video flow by virtue of the system’s adaptation functionality:

$$[(\lambda, \lambda'), T_{\text{cAIMD}}, \frac{d\lambda'}{dt}, (\Delta\lambda', T_l)],$$

where λ is the desired send rate, λ' is the sustainable send rate, $\Delta\lambda'$ is the rate swing in the steady-state and T_l is the corresponding limit-cycle time, $\frac{d\lambda'}{dt}$ depicts how fast the rate fluctuates in the limit-cycle region, and T_{cAIMD} is the rate convergence time to effect a congestion relief (i.e., the time to move from a congested state to the steady-state) using the AIMD rule. Rate jitter is a key QoE parameter associated with the steady-state behavior, and may be captured in terms of the range and frequency of rate fluctuations $[\frac{d\lambda'}{dt}, (\Delta\lambda', T_l)]$. The convergence latency T_{cAIMD} depicts how responsive is the system to a congestion (which is a QoE indicator).

The BAVT system exhibits a non-linear relationship between B_{av}^* and the QoE parameters: rate convergence and rate jitter. Figures 3-4 show the experimental results for a converse version of AIMD algorithm that allocates/de-allocates bandwidth in incremental steps to sustain a given flow rate. In this converse version, the roles of β and α respectively are to determine the additional bandwidth allocation needed to counter an observed packet loss and the bandwidth de-allocation margin in a case of no packet loss. An initial traffic load of 10 video flows (from *Jurassic Park* clips) for an allocation of 30 *mps* was increased to 15 flows, with the loss-tolerance set as: $\delta_l = 0.05\%$ and $\delta_h = 0.08\%$. The final steady-state allocation determined by the algorithm is in the range 45-46 *mps*. Results on the adaptation performance in reaching the steady-state illuminate two points of interest. First, the rate convergence time T_c varies with the β -parameter: $T_c = 3.25$ *sec* for $(\beta = 0.55, \alpha = 0.012)$ and $T_c = 2.1$ *sec* for $(\beta = 0.65, \alpha = 0.012)$. Second, the limit-cycle time in the steady-state are about 0.9 *sec* and 0.75 *sec* respectively. These results indicate that the ability to dynamically adjust the AIMD parameters as the BAVT operating region changes is highly desirable to improve the human-oriented QoS aspects (i.e., QoE).

The rate adaptation control module vis-a-vis the QoE management, and the underlying adaptation elements form the core parts of BAVT system. The degree of control exercised on the rate adaptation module may vary from one system to another. But the core adaptation functions are fixed, with a suitable management and signaling interface to make them *programmable* (in a functional sense) by having an external module supply the QoE-oriented user-level parameters and algorithm-oriented AIMD parameters⁴.

References

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⁴The level of rate sustainability $(\lambda - \lambda')$ for a flow depicts how much the system can offer relative to the flow's expectation. Regardless of the type of rate adjustments employed, $(\lambda - \lambda')$ depends only on the available bandwidth B_{av}^* along the transport path — which is outside the purview of a rate adjustment algorithm. Thus, treating $(\lambda - \lambda')$ as an adjustable QoE parameter involves exercising control on B_{av}^* itself: say, adding high-capacity links along the path and/or lowering the cross-traffic.

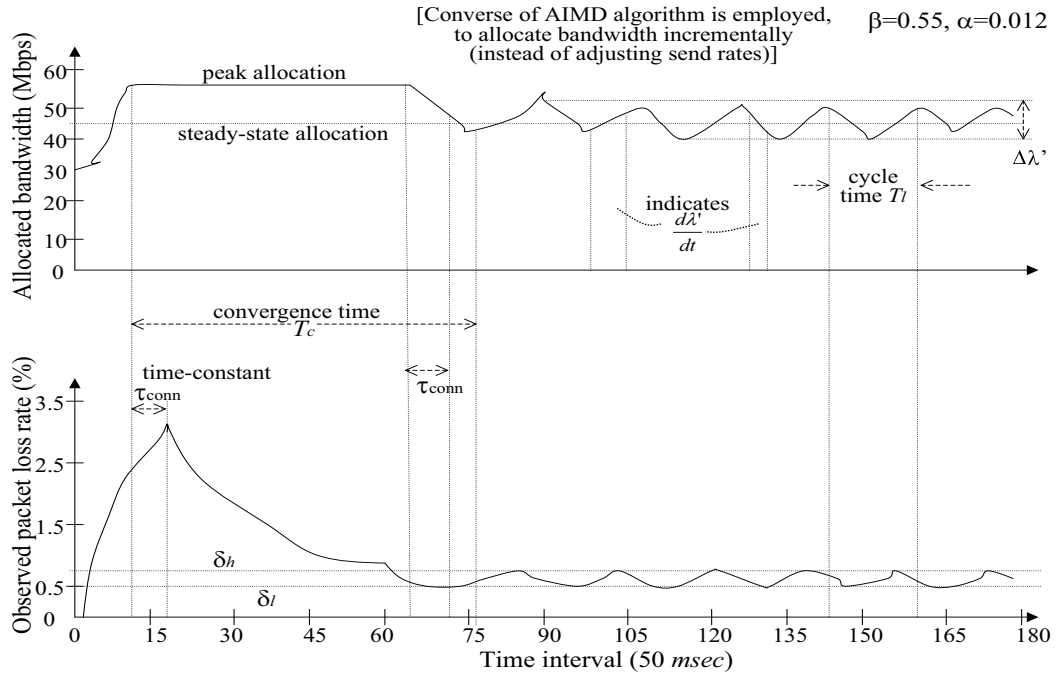


Figure 3: Adaptation behavior of BAVT system with AIMD parameters: $\beta = 0.55, \alpha = 0.012$

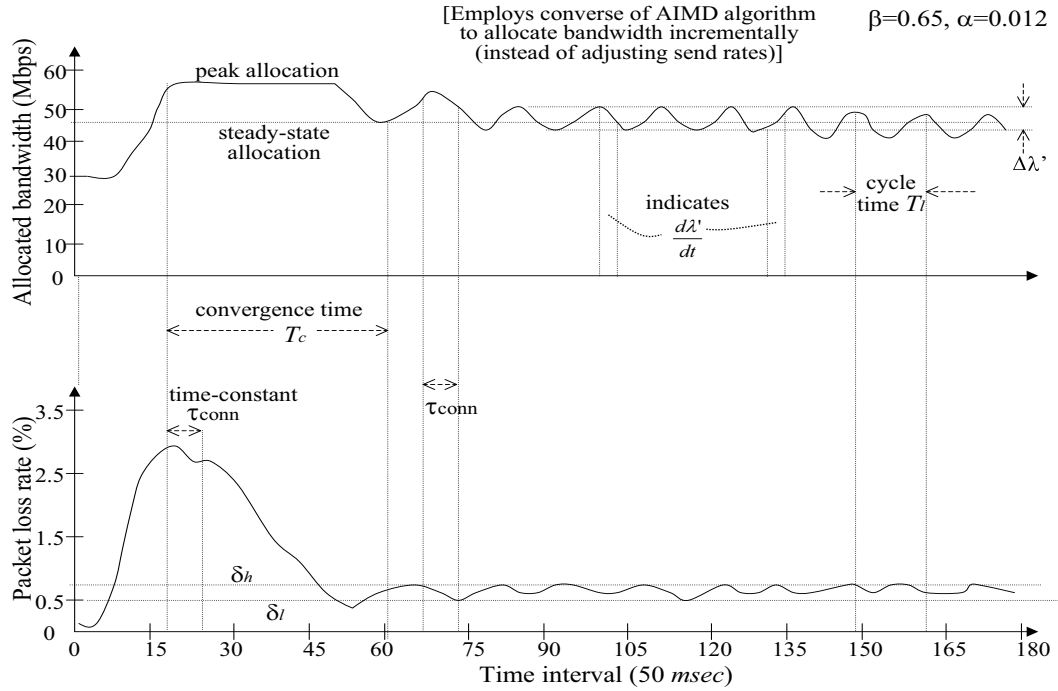


Figure 4: Adaptation behavior of BAVT system with AIMD parameters: $\beta = 0.65, \alpha = 0.012$